Investigating the Role of Olfaction in Female Cichlid Reproductive Behavior

Cade McGeehan, Ofure Osunbor, Tahir Haroon, Ariel Bazan, Vijay Jayasuriya, William Lamousin, Jason Liu, Charles Phan, Shubh Sharma, Bhargav Srinivasan

[1]Scott Juntti, [1]Coltan G. Parker

[1]Department of Biology, University of Maryland, College Park

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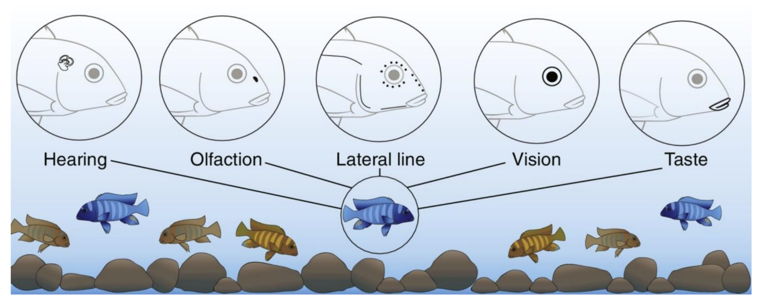
Background

Cichlids (*Astatotilapia burtoni*)

- ~2000 species located primarily in East African Lakes[1]
- Evolved diverse morphologies and behaviors among species, making them a strong model species
- Females display dull coloration while males are vibrant
- Rely on five sensory systems (hearing, mechanosensation via a lateral line, taste, olfaction, vision) to guide behavior[1]
- Reliance on these sensory systems is dynamically adjusted depending on environmental conditions and behavioral demands

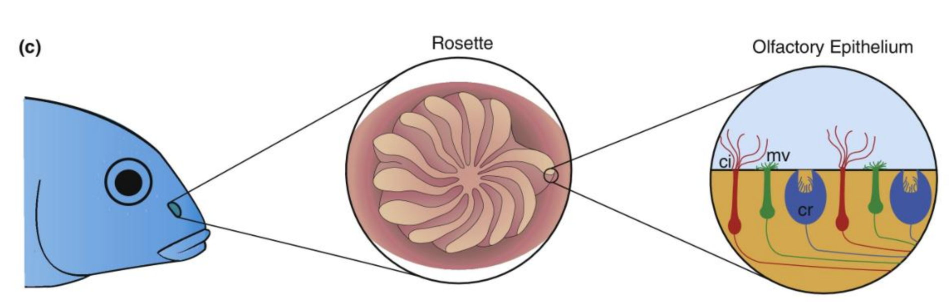


Cichlid Sensory Systems [1]

male *Astatotilapia burtoni* [4]

Olfactory System

- Odorants bind to specific chemosensory receptors in the olfactory epithelium:
  - Vomeronasal type 1 and 2 receptors (V1Rs and V2Rs)[5]
  - Olfactory receptors (ORs)[5]
  - Trace amine associated receptors (TAARs)[5]
- Chemosensory receptors are differentially expressed across olfactory sensory neurons (OSNs): ciliated, microvillous, and crypt neurons.
- Binding triggers a signaling cascade that converts chemical cues into electrical signals.
- These signals are transmitted to the brain providing information about the environment influencing survival and social behaviors
- Variation in OSN abundance and receptor composition likely drives functional specializations, shaping olfactory processing and potentially influencing behavior



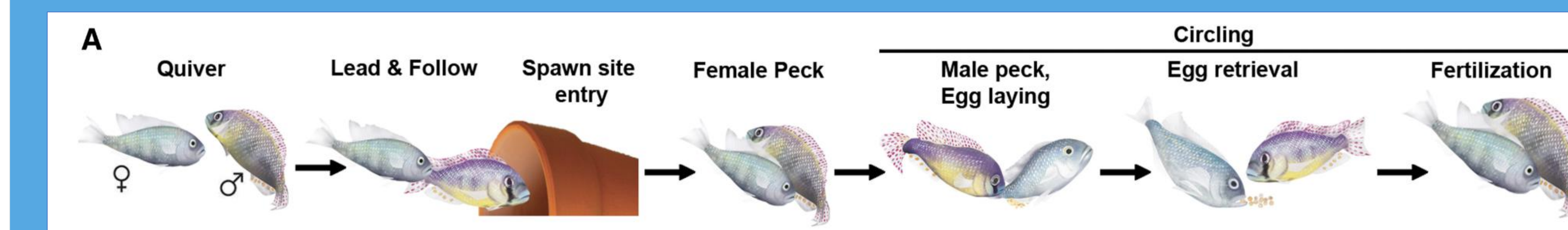
Cichlid Olfactory Anatomy[1]

Knowledge Gap

- Distinct olfactory neuron pathways express different ion channels, but individual contributors to the behavior are not fully understood
- Current annotation approaches rely on manual behavioral scoring, limiting efficient quantitative comparisons across experimental conditions
- We aim to:
  - Determine whether olfaction is necessary for reproductive behavior
  - Identify which olfactory pathways contribute to quantified reproductive behavior

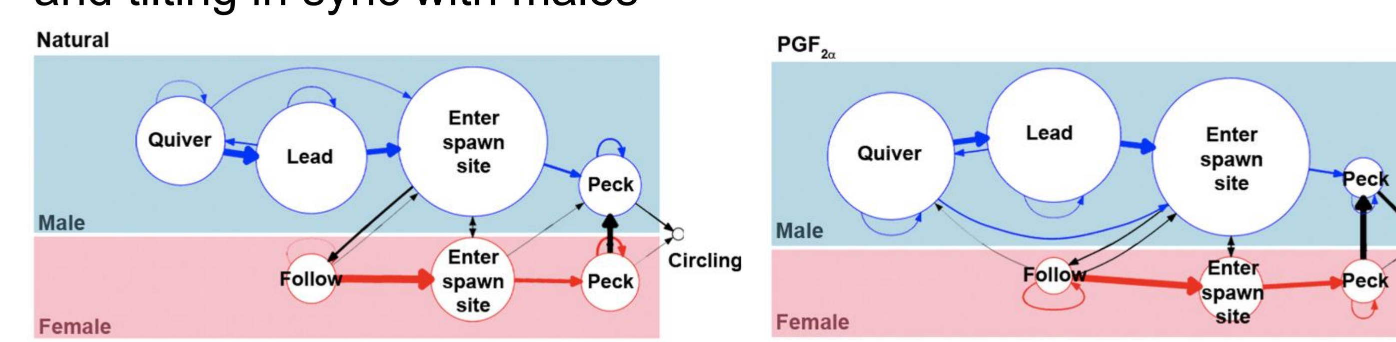
SAM3 computer vision model masks an A. Burtoni fish

Reproductive Behaviors



Steps in the mating behavior process[4]

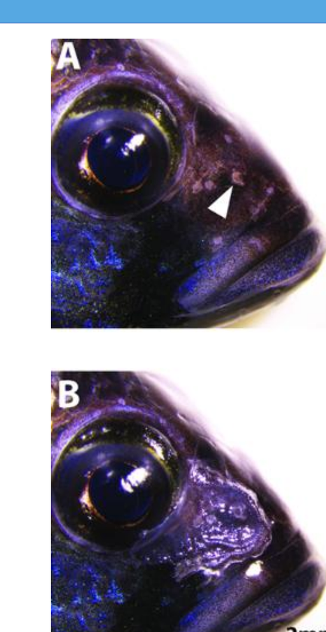
- Male and female cichlids exhibit distinct behaviors during courtship and spawning.[4]
- Males display bright coloration, fins, and body quivering to attract females.
- Males engage in pecking, tilting behavior, and leading to lure female mates.[4]
- Females reciprocate by following male leads and display pecking and tilting in sync with males[4]



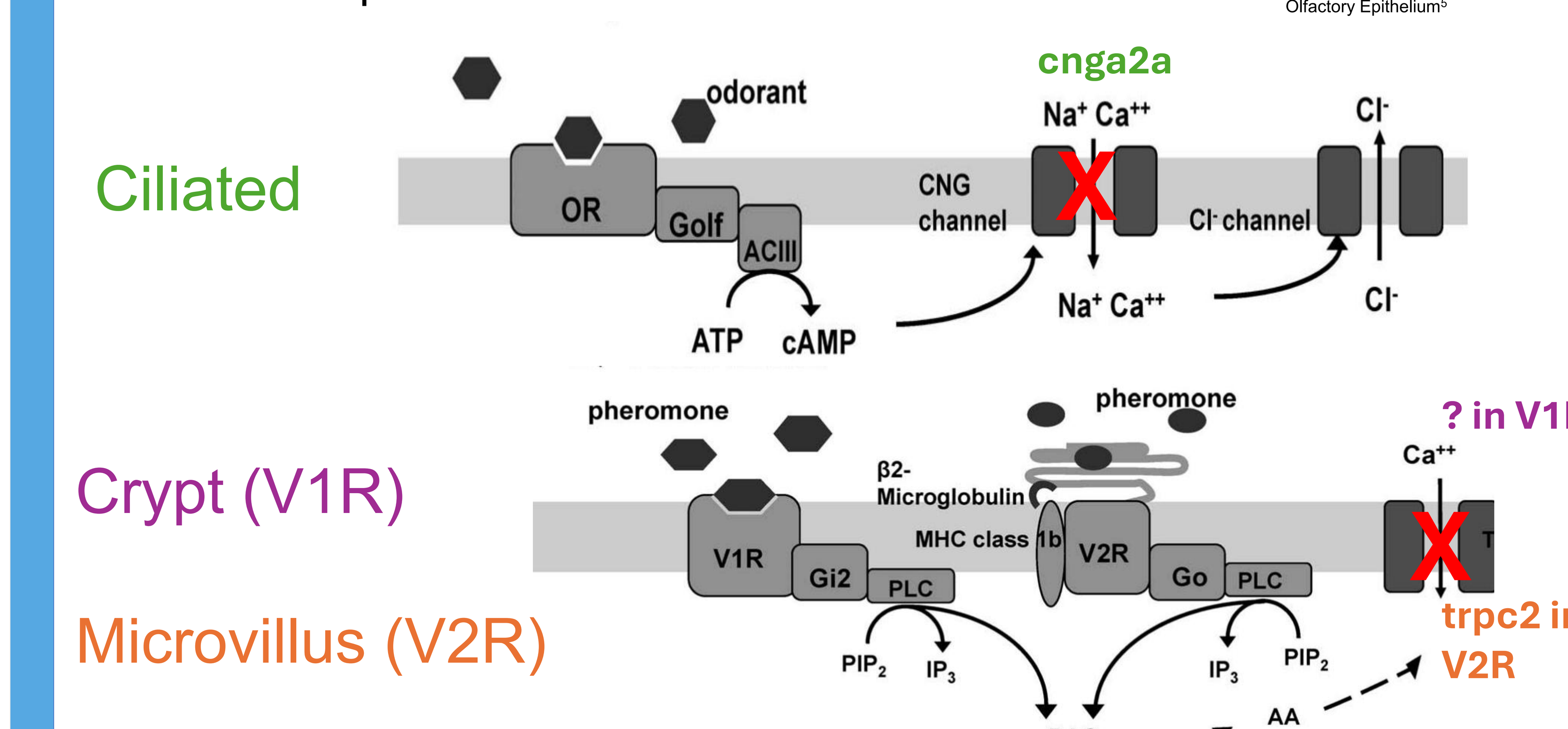
Natural (left) mating behaviors of the male (top) and female(bottom) cichlid fish. PGF2a altered (right) mating behaviors of male and female cichlid fish.[4]

Strategy for Determining Which Olfactory Cells Drive Mating

1. Block the nose of wild type females eliminating their olfactory system
2. Compare frequency of behaviors to control females to identify if olfaction plays a role in female reproductive behaviors
3. Mutate each OSN specific ion channel, silencing one OSN type at a time
4. Annotate and compare frequency of reproductive behaviors of mutants to nose blocked females
5. Identify if any of the OSN play a significant role in female reproductive behavior



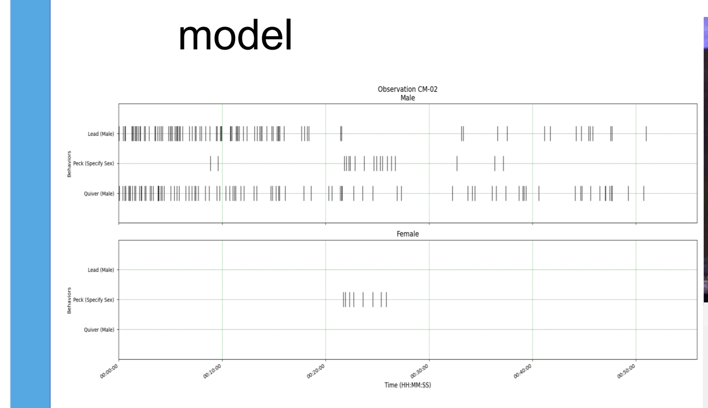
Nasal Block for Silencing the Olfactory Epithelium[5]



Signal transduction pathways of olfactory neurons[3]

Behavioral Quantification System for Analyzing Behavioral Changes

- Annotate experimental videos using BORIS software
- Assess inter-rater reliability scores across multiple annotators in hopes of high agreement
- Generated annotated datasets used to train a computer vision model



Plot of observation displays timestamps of different annotations across the video.

	CM-02	CP-01	TH-02
CM-02	1	0.635	0.677
CP-01	0.635	1	0.634
TH-02	0.677	0.634	1

Inter-rater reliability scores help compare how close individuals' annotations are to one another

Ethogram

	Key	Code	Type	Description
1	q	Quiver (Male)	Point event	Male displays it...
2	l	Lead (Male)	Point event	Male leads ...
3	f	Follow (Female)	Point event	Female follows ...
4	p	Peck (Specify ...	Point event	Either fish peck...
5	n	In Nest (Specify...	State event	Fish has the ...
6	b	Bite (Male)	Point event	Male nips/bites...
7	c	Chase/Charge ...	Point event	Male charges at...
8	r	Run/Flee ...	Point event	Female flees ...
9	o	Circle	Point event	Log any time a ...
10	t	Tilt (Specify Sex)	Point event	Either fish tilts/...

BORIS interface used for annotating behaviors

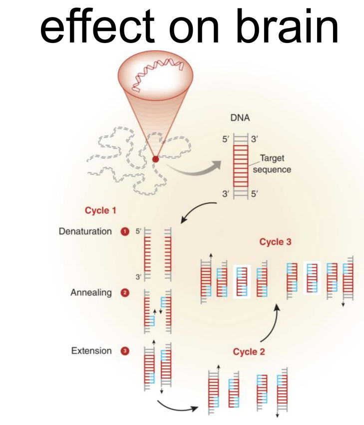
Comparative Techniques

Genotyping

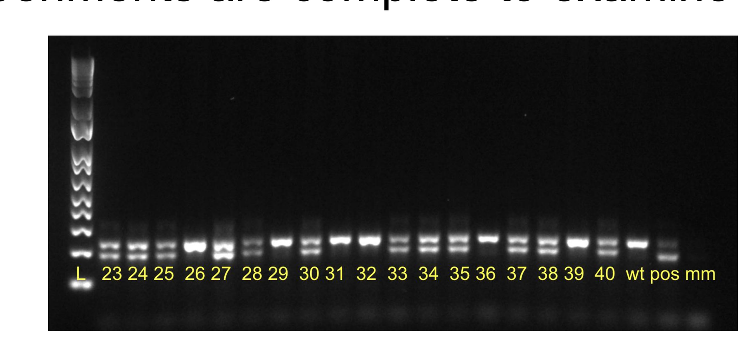
- Collect fin clips to isolate DNA
- Amplify target DNA using PCR
- Run PCR products on an electrophoresis gel
- Compare band patterns to controls to identify individuals with the mutant allele

Hormone injections

- Injecting PGF2a hormone in female cichlids while simultaneously blocking their nose
- Compare reproductive behavior frequency of nose blocked females to control females
- Dissections once experiments are complete to examine effect on brain



Polymerase Chain Reaction (PCR) for DNA Amplification[2]



Gel electrophoresis of cichlid DNA (23–40) with controls distinguishing mutant (two bands) from wild type genotypes (one band)

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Developing a Computer Vision Model for Cichlid Behavior Annotation

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Bhargav Srinivasan, William Lamousin, Charles Phan, Jason Liu, Shubh Sharma, Vijay Jayasuriya, Ariel Bazan, Cade McGeehan, Ofure Osunbor, Tahir Haroon

Pulkit Kumar, Abhinav Shrivastava

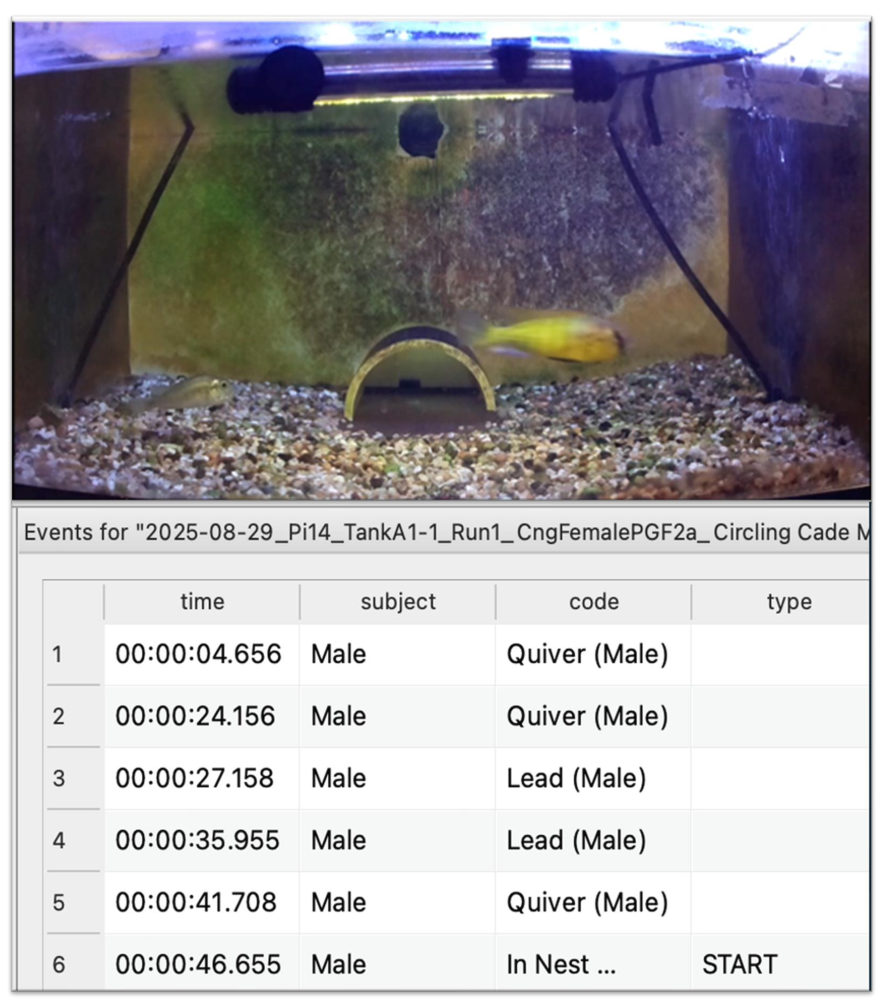
GEMSTONE
Honors College
University of Maryland

Goal

Leverage computer vision to automate annotations for reproductive behavior in fish

Motivation

- Behavioral studies require **tedious manual annotation** of **repetitive videos**
- Some behaviors occur **once per month**
- Modern computer vision techniques make these practices **obsolete**



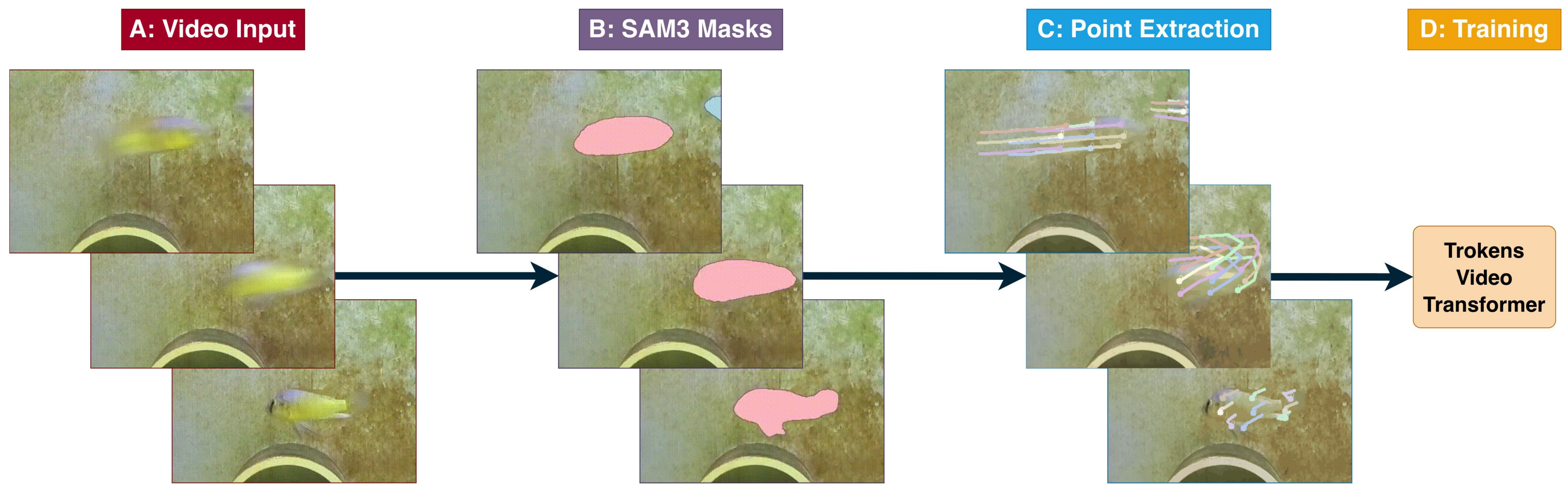
BORIS: Manual annotation software

SLEAP: Pose tracking software

Our Solution

- **Computer Vision Pipeline** – automated annotation tool
- **Deep Learning Few-Shot Framework** – leverage thousands of hours of prerecorded videos

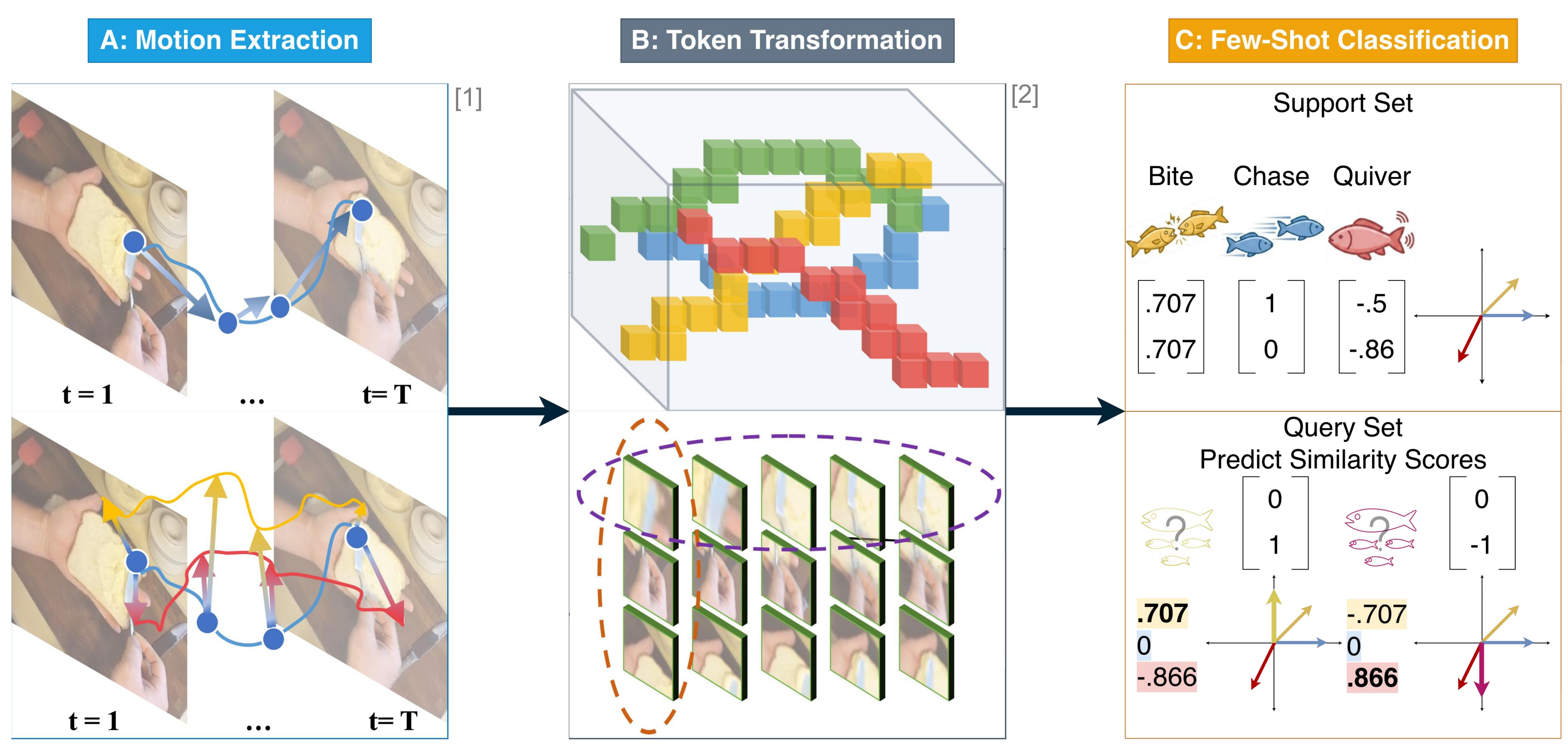
Our Method



Our Method, Trokens++ gains spatial context by extracting features from Segment Anything Model 3

(A) 3-10 second short clips are segmented from 1-2 hour large clips to provide full context of a behavior
(B) Given a video input, we extract and propagate the masks of at most two fish using SAM3
(C) We extract notable points from these masks using a grid-based sampling method
(D) The clips are run through the Trokens video transformer described below to train the model

Trokens Video Transformer



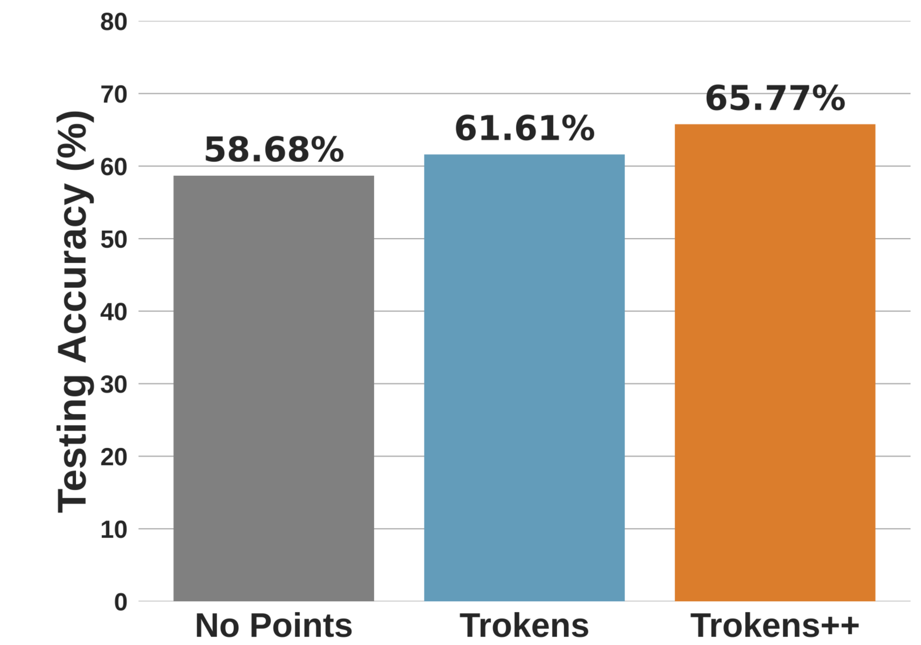
The Trokens Model uses a few-shot action recognition pipeline to learn classification of fish behavior

(A) After pre-processing, we use the trajectories to create semantic-aware relational trajectory tokens
(B) These tokens are fed into a Decoupled Space-Time Transformer to create embeddings
(C) The model then predicts query video classes based off similarity scores to given support videos

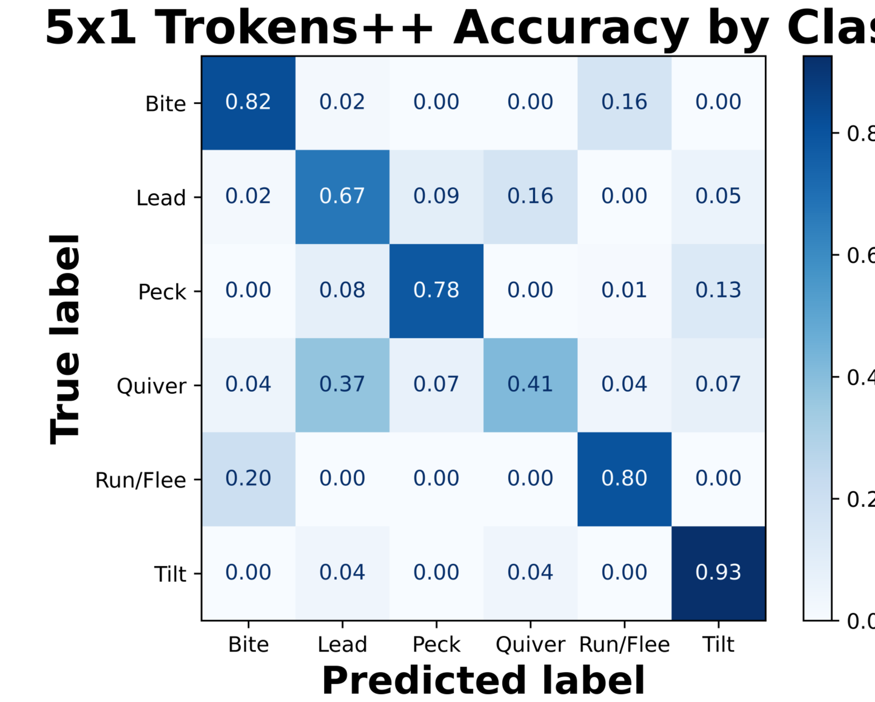
Figures from: [1] Trokens: Semantic-Aware Relational Trajectory Tokens for Few-Shot Action Recognition
[2] Trajectory-aligned Space-time Tokens for Few-shot Action Recognition

Results

- 5-way 1-shot Trokens++ model achieved the greatest final accuracy



5x1 Trokens++ Accuracy by Class

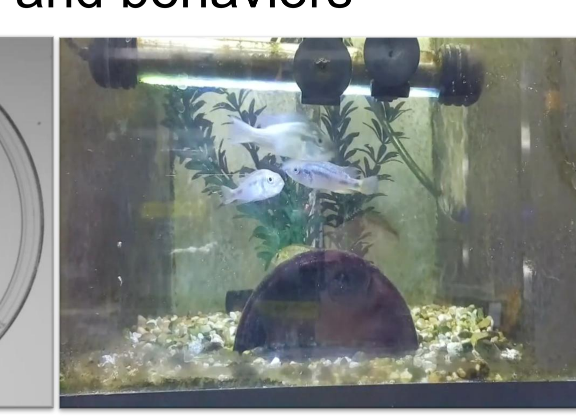


Models struggled to classify quiver and lead behaviors

Future Plans

- **Integration** into full video analysis pipeline
- **Generalize model** to other species, lab environments, and behaviors

Cichlid Fry Interactions
Angueyra Lab



Multiple Adult Interactions
Juntti Lab

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